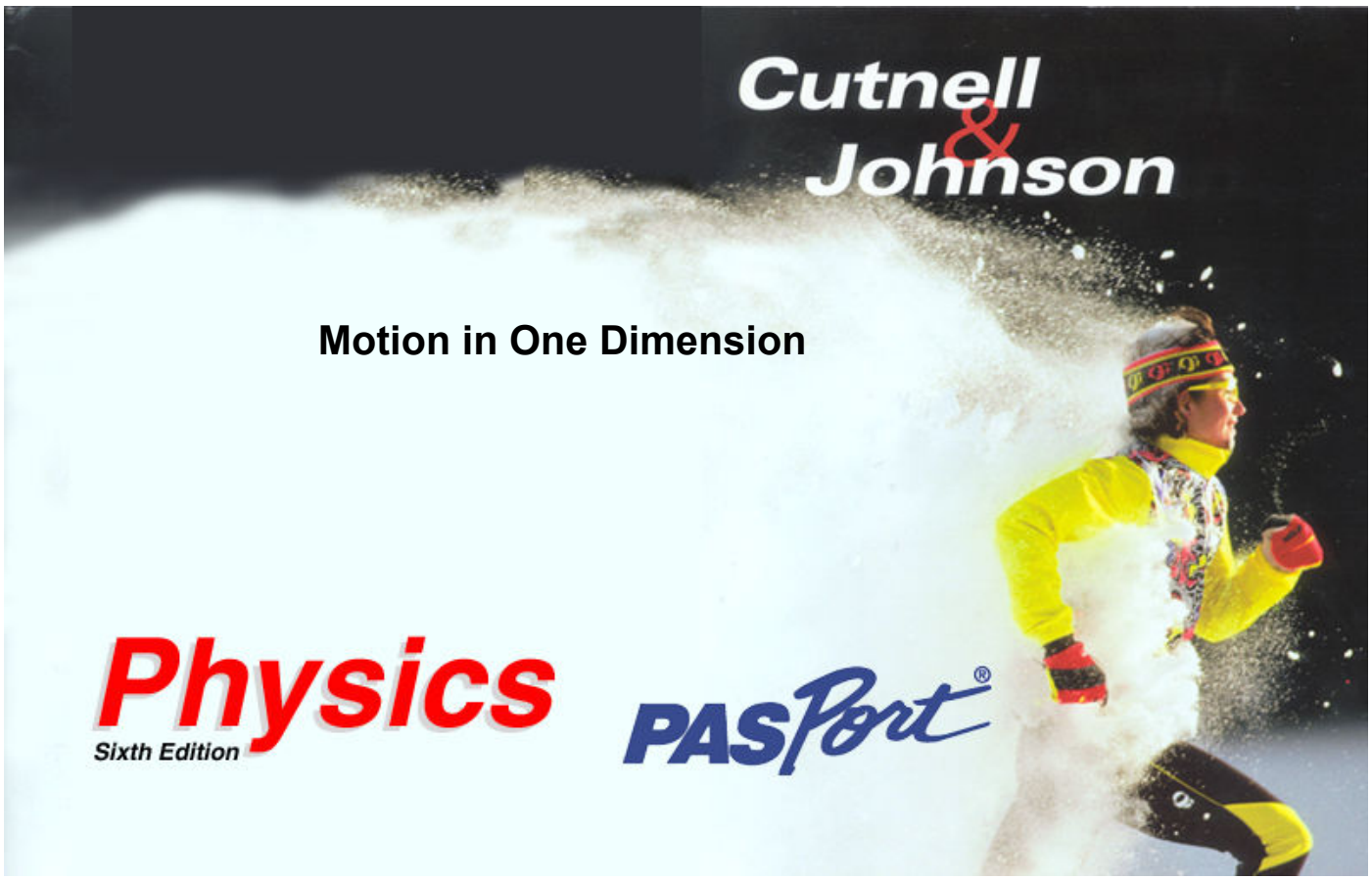

Cutnell & Johnson
Motion in One Dimension
Physics
Sixth Edition
PASPort®

Student Information

§ Enter the appropriate information in the boxes below.

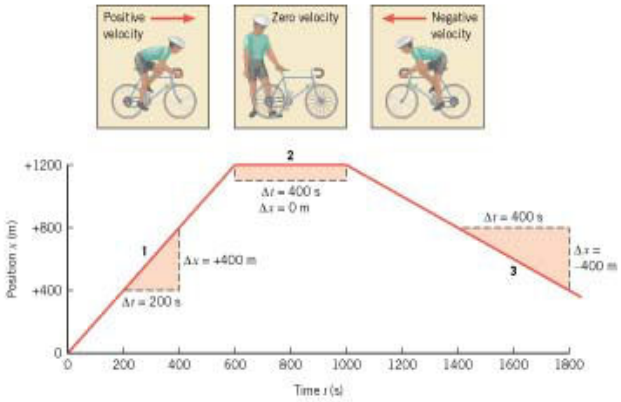
Date:	
Your Name:	
Group:	
Period:	

Introduction

Exploration Goals

§ The purpose of this activity is to explore graphs of motion (position versus time and velocity versus time). Use a Motion Sensor to measure *your* motion as you move back and forth relative to the sensor in a straight line at different speeds.

§ The challenge is to move in such a way that a plot of your motion will "match" the position-time and velocity-time graphs that are provided for you. Examine your motion using the graphs. The goal is to build an understanding of the relationships between position and velocity.



Learning Outcomes

You will be able to:

- Understand the relationship between position and velocity.
- Understand that velocity is the rate of change of position over time.
- Relate the physical motion of an object to graphical representations of the motion.
- Use the software to determine how well your motion "matches" the pre-defined plots of position and velocity on the Graph display.
- Understand the idea that all motion occurs within a frame of reference.

Hypothesis

§ What is the relationship between the motion of an object and the graphs which represent its position versus time and velocity versus time?

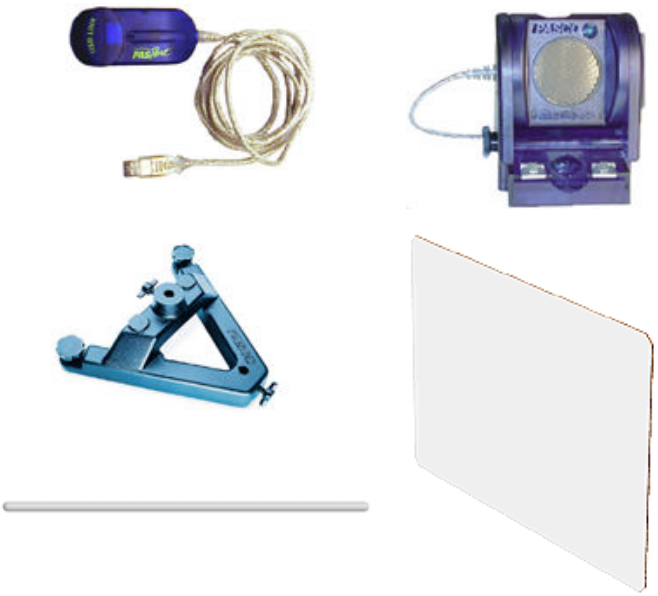
Background

§ When describing the motion of an object, knowing where it is relative to a reference point, how fast and in what direction it is moving, and how it is accelerating (changing its rate of motion) is essential. A sonar ranging device such as the PASPORT Motion Sensor uses pulses of ultrasound that reflect from an object to determine the position of the object. As the object moves, the change in its position is measured many times each second. The change in position from moment to moment is expressed as a velocity (meters per second). The change in velocity from moment to moment is expressed as an acceleration (meters per second per second). The position of an object at a particular time can be plotted on a graph. You can also graph the velocity and acceleration of the object versus time. A graph is a mathematical picture of the motion of an object. For this reason, it is important to understand how to interpret a graph of position, velocity, or acceleration versus time. In this activity you will plot a graph of position in real-time, that is, as the motion is happening.

» For more information see Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 2, Sections 2.1 to 2.3.

Materials

- PASPORT USB Link
- PASPORT Motion Sensor
- Large Rod Base
- Support Rod
- Motion Sensor Reflector Board

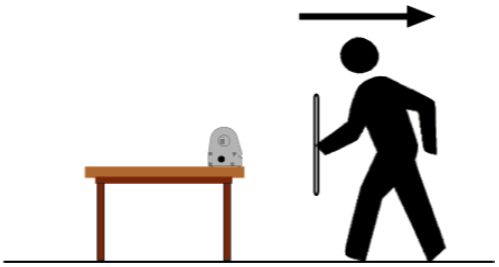


Setup *Equipment Setup*

§ Mount the Motion Sensor on a support rod so that it is aimed at your midsection when you are standing in front of the sensor. Make sure that you can move at least 2 meters away from the Motion Sensor.

§ Position the computer monitor so you can see the screen while you move away from the Motion Sensor.

Safety Reminder: You will be moving backwards for part of



Process *Prepare to Record Your Position*

Read this page before you take data. The next page is where you will record data.

§ When you are ready, stand in front of the Motion Sensor.

§ When everything is ready, start recording data.

» There is a three-second countdown before data recording begins. A "pointer" on the vertical axis of the graph will move up and down as you move forward and backward relative to the sensor. Use the feedback from the "pointer" to adjust your start position.

» **WARNING:** You will be moving backward, so clear the area behind you.

» The Motion Sensor makes a faint clicking noise.

§ Watch the plot of your motion on the graph and try to move so that the plot of your motion matches the Position versus Time plot already there.

» The digits display shows how closely you matched the graph. The closer to 100, the better.

Process *Record Your Position*

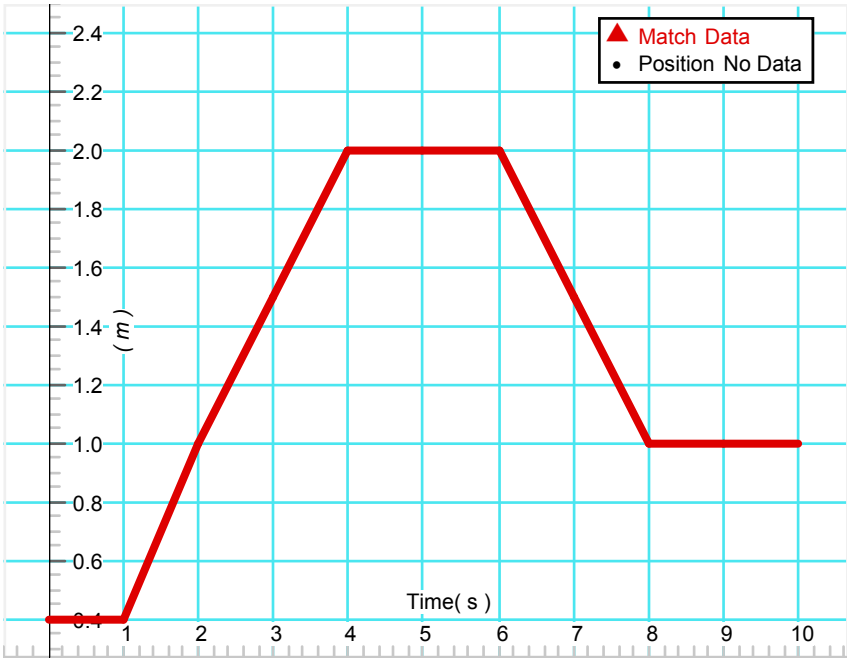
§ Click **Start**.

§ Try to move so the plot of your motion matches the Position versus Time plot already there.

§ Recording will **Stop** automatically at 10 sec.

§ Repeat the data recording process a second and a third time. Try to improve the match.

• MatchScore No Data



Process *Prepare to Record Your Velocity*

You will now do the same procedure using a Velocity vs. Time graph on the next page.

§ When you are ready, stand in front of the Motion Sensor.

» **WARNING**:You will be moving backward, so clear the area behind you.

§ When everything is ready, start recording data.

» There is a three-second countdown before data recording begins. A "pointer" on the vertical axis of the graph will move up and down as you move forward and backward relative to the sensor. Use the feedback from the "pointer" to adjust your start position.

» The Motion Sensor will make a faint clicking noise.

§ Watch the plot of your motion on the Graph and try to move so the plot of your motion matches the Velocity versus Time plot already there.

» The digits display shows how closely you matched the graph. The closer to 100, the better.

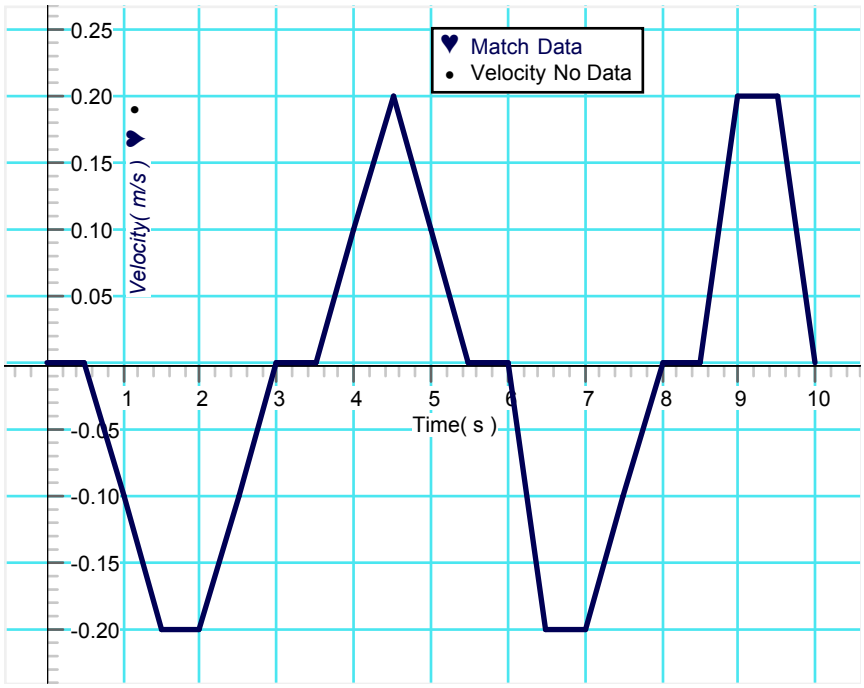
Process *Record Your Velocity*

§ Click **Start**.

§ Try to move so the plot of your motion matches the Velocity versus Time plot already there.

§ Recording will **Stop** automatically at 10 s.

§ Repeat the data recording process a second and a third time. Try to improve the match.



♣ MatchVelocity Run #2

91.43

Observations

§ How well did your motion graphs match the provided graphs?

§ What was the meaning of the part of the position plot where the slope was positive (upward)?

§ What was the meaning of the part of the velocity plot where the slope was positive (upward)?

Analyze *Variables and Controls*

§ What were the variables in this activity?

§ Which of those did you control?

§ Which did you manipulate and what was the response?

Synthesize *Error Analysis*

§ Was the plot of your motion as smooth as the provided graph?

§ If not, why was your graph not as smooth as the provided one?

Synthesize *Conclusions*

§ What is the description of your motion for both graphs? (Example: "Constant speed for 2 seconds followed by no motion for 3 seconds, etc.")

» Position versus Time

» Velocity versus Time

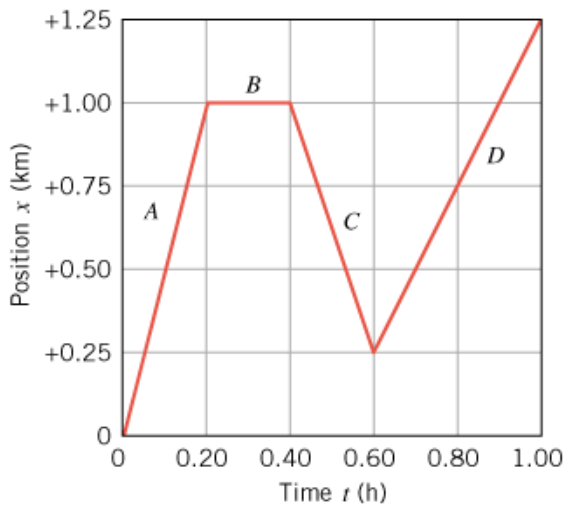
Synthesize *Applications*

§ What are some physical phenomena where knowing your motion is important?

Synthesize *Extension Problem*

§ In Cutnell & Johnson, Physics, 6th ed., Volume One, Chapter 2, Problem 60, page 52.

A person who walks for exercise produces the position-time graph given with this problem. (a) Without doing any calculations, decide which segments of the graph (*A*, *B*, *C*, or *D*) indicate positive, negative, and zero average velocities. (b) Calculate the average velocity for each segment to verify your answers to part (a).



Note: Put the answer on the next page.

Synthesize *Extension Problem Answer*

» Answer for the extension problem

Notes